MILITARY SPECIFICATION

TEST SET, RADAR, AN/APM-72

This specification has been approved by the Bureau of Aeronautics, Department of the Navy.

1. SCOPE AND CLASSIFICATION

1.1 Scope - The equipment covered by this specification shall be designed for use in preflight checking, adjusting, and aligning the Radar Set AN/APG-30.

1.2 Classification - The Radar Test Set AN/APM-72 shall be of one type and shall consist of:

Nomenclature	Type Designation	Max Allow Size (in.) L	W	H	Max Allow Wt. (lb)	Applicable Paragraph
Test Set, Radar	TS-787(*)/APM-72	17	13	12-7/8	40	3.3.1
Box, Interconnecting	J-565(*)/APM-72	8	6	3-1/2	8	3.3.2
Cable, Assembly	CX-2452/U	36				3.3.2.5
Cable, Assembly	CX-2451/U	120			3.5	3.3.3
Cable Assembly, RF	CG-572/U	156			3	3.3.4
ANL Probe						3.3.5
Video Adapter						3.3.6
Case, Transit	CY ( )	26	16	16	30	3.3.7

* The parenthesis in the type designation shall be replaced by a capital letter furnished by the procuring agency upon application by the contractor in accordance with the requirements of paragraph 3.2.12 after each contract has been awarded. The complete type number shall be used on nameplates, shipping records, and instruction books as applicable.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, shall form a part of this specification to the extent specified herein:

2.2 Specification

Military

MIL-T-27	Transformers and Inductors (Audio, Power, and Pulse)
MIL-T-945	Test Equipment, For Use With Electronic Equipment; General Specification
MIL-C-3608	Connectors, BNC, for Radio Frequency Cables
MIL-I-6181	Interference Limits and Tests; Aircraft Electrical and Electronic Equipment
JAN-C-62	Capacitors, Dry-Electrolytic, Polarized
JAN-T-152	Treatment, Moisture-and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment; General Process For
JAN-P-658	Packaging and Packing of Electrical Equipment and Spare Parts (Electronic, Electrical, and Electromechanical)

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MIL-N-18307	Nomenclature and Nameplates for Electronic and Associated Equipment

Standards

MIL-STD-129	Marking for Shipment and Storage

Drawings

Air Force-Navy Aeronautical

AN3102	Connectors; Electrical, Receptacle, Box Mounting
AN3106	Connectors; Electrical, Plugs, Straight
AN3436	Terminal Block Assembly; Electrical

Naval Aircraft Factory

1128	Box; Junction, Radio and Electrical System, Conduit

2.3 Availability of Applicable Documents

(1) When requesting applicable documents, refer to both title and number. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Air Station, Johnsville, Pennsylvania, Attn: Technical Records Division.

2.4 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring agency.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials

3.1.1 General - The equipment shall fulfill the following requirements for parts and materials:

(1) Design Objectives - Minimum size and weight, simplicity of operation, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives in the design of the equipment specified herein. The use of materials, parts, and processes other than those required by Specification MIL-T-945 shall be investigated, and when it appears that a substantial reduction in size and weight or improvement in simplicity of design, performance, or reliability can be realized by their substitution, a request for approval of a nonstandard part shall be submitted to the procuring agency for consideration. Each request shall be accompanied by complete supporting information.

(2) Selection of Parts and Materials - The fulfillment of major design objectives shall govern the selection of parts and materials. All parts and materials shall meet the requirements of Specification MIL-T-945; however, if the contractor can demonstrate that the use of such parts

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and materials will not fulfill the major design objectives, then parts and materials shall be used which most nearly meet or exceed the requirements of SpecificationMIL-T-945.

(3) Approval for Nonstandard Parts and Materials - Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-T-945. Requests for approval must be forwarded to the procuring agency by the contractor before parts are purchased or fabricated.

(4) Workmanship - Workmanship, fabrication, and manufacturing process shall conform to the requirements of Specification MIL-T-945.

(5) Transformers, Inductors, and Capacitors - The use of open-type transformers and inductors will be acceptable if such units are protected by encapsulating material approved by the procuring agency. The Grade 2 requirements of Specification MIL-T-27 shall apply to the open-type units only to the extent that these requirements be fulfilled by the Flexseal, Fosterite, and Permafil processes, or equivalent. Electrolytic capacitors may be used if they meet the requirements of Specification JAN-C-62.

(6) Electron Tubes - Electron tubes used in the equipment shall be chosen in accordance with the requirements of Specification MIL-T-945.

3.2 General Design Requirements

3.2.1 General - This equipment shall fulfill all applicable design and performance requirements of Specification MIL-T-945 unless otherwise specified herein.

3.2.2 Form Factor and Weight - The over-all dimensions of the equipment shall not be greater than 17 inches long, 13 inches wide, and 12-7/8 inches high. The total weight of the entire equipment shall not be greater than 85 pounds.

3.2.3 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions, or reasonable combinations of service conditions, as specified in MIL-T-945 except that the requirements specified shall apply only within the temperature range from -40° to +55° C.

3.2.4 Operating Power - The equipment shall operate as specified herein from external 115V ± 10 per cent, 320 to 1760 cps., a-c power source and d-c power supplied by Radar Set AN/APG-30. The a-c power required shall not be greater than 130 volt- amperes. The d-c power required shall not be greater than 42 watts at +28 volts.

3.2.5 Operating Life

3.2.5.1 Reliable Operating Life - The equipment shall be capable of operation, as specified herein, for a period of 200 hours without requiring servicing or replacement of parts. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.5.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing or replacement of parts. The life of such parts and parts requiring replacement within this interval shall be specified by the manufacturer.

3.2.6 Operational Stability - The equipment shall operate with optimum performance for at least 200 hours, continuously or intermittently, without necessity for adjustment of controls other than those located on the front panel.

3.2.7 Warm-Up - The following warm-up periods shall not exceed 20 minutes for ambient temperatures above 0° C, and 30 minutes for ambient temperatures below 0° C. At the termination of the warm-up period, all other requirements of this specification shall apply, with no readjustment of internal controls.

3.2.8 Interference - The equipment shall fulfill the interference requirements of Specification MIL-I-6181.

3.2.9 Cables and Connectors - Cables and connectors shall be in accordance with the requirements of Specification MIL-T-945. Insofar as is practical, all front panel connectors shall be mounted along the lower

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edge of the front panel and in no case shall their location be such that the connecting cables will interfere with the operating controls.

3.2.10 Fungus and Moisture Treatment - The equipment shall be moisture and fungus-proofed by selection of parts and materials that are non-nutrient for fungus. The parts and materials shall be treated so that prior to use in the equipment over-all spraying of the equipment is not necessary.

3.2.11 Interchangeability - Mechanical and electrical interchangeability shall exist between similar assemblies, subassemblies, and replaceable parts regardless of the manufacturer or supplier. Interchangeability for the purposes of this paragraph does not mean identity, but requires that a substitution of such like assemblies, subassemblies, and replaceable parts be easily effected without physical or electrical modifications to any part of the equipment including cabling, wiring, and mounting. In the design of the equipment, provisions shall be made for design tolerance to accommodate various articles such as tubes, resistors, and other components having the limiting dimensions and characteristics set forth in the specifications for the particular component involved without departure from the specified performance.

3.2.12 Nameplates and Nomenclature - Nameplates and nomenclature shall be in accordance with the requirements of Specification MIL-N-18307.

3.3 Detail Design Requirements - The Test Set AN/APM-72 shall provide means for checking and adjusting the following AN/APG-30 radar system parameters:

Aircraft d-c supply voltage
Radar d-c supply voltages
Magnetron current
Radar afc crystal current
On-target Lamp
Radar a-c regulator output voltage and frequency
Variable frequency line voltage and frequency
Radar repetition rate
Radar preamplifier crystal current
Radar range calibration
Servo range calibration
Radar range tracking
Automatic Noise Leveler (ANL) voltage
Radar video output
Radar Gates - Out Function

3.3.1 Test Set, TS-787(*)/APM-72 - The TS-787(*)/APM-72 shall be designed to meet the following requirements:

3.3.1.1 Form Factor and Weights - The dimensions of the TS-787(*)/APM-72 shall not be greater than 17 inches long, 13 inches wide and 12-7/8 inches high, and the weight shall not be greater than 40 pounds.

3.3.1.2 Contents of Unit - The Test Set TS-787(*)/APM-72 shall consist of a precision bridge circuit, a marker, a tracking circuit, an "A" scope circuit, and the necessary controls and meters.

3.3.1.2.1 Bridge Circuit - The bridge circuit shall provide a means of balancing or comparing a standard voltage from a calibrated voltage divider against the range voltage output of the AN/APG-30.

(1) Power Calibration - The deviation in calibration shall not be greater than ±0.5 V as indicated by the DC Volts meter when the FUNCTION SELECTOR is set at CAL, and the DC VOLTS control is set at +200 REF, when such a calibration check is conducted after any functional operation is made with the equipment.

(2) Balance - The YARDS ERROR meter shall read 50 ± 2 yards when the BALANCE meter is on the mark and the BRIDGE BAL control is turned full clockwise. Suitable adjustment shall be provided so that an exact 50-yard indication may be obtained when the BALANCE meter is balanced and the FINE BAL switch is engaged.

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(3) Range Zero Set - With the FUNCTION SELECTOR at ANL and RANGE SELECTOR at 3900, the range meter reading shall be adjustable from 3000 to better than 4500 feet.

(4) Range Accuracy - After the range has been zeroed at 3900 ft, the RANGE meter shall indicate any of the settings of the RANGE SELECTOR within±200 feet.

(5) Yards Error Accuracy - Under a fine balanced system, the YARDS ERROR meter shall not indicate any error greater than:

Range (ft)	± Error (yd)
900	1.5
2250	2.5
2400	2.5
3900	4.0
5400	5.5
8400	11.0

(6) Servo Control - With the FUNCTION SELECTOR at SERVO, the RANGE SELECTOR at any position, and fine balance depressed, a servo control shall be provided to give one-half full scale BALANCE indications to either side of the balanced mark.

3.3.1.2.2 Pulse Delay Circuit - A calibrated pulse delay circuit shall be provided to receive trigger pulses from the AN/APG-30 transmitter and reproduce them as delayed pulses to simulate in one position of the MARKER SELECTOR switch, target returns from 900 and 2250 feet or, in the second position target returns from 900 and 5400 feet. When the MARKER SELECTOR Switch is in the TRACK position, an uncalibrated delayed pulse may be obtained which may be varied, by rotating the TRACK RANGE control to simulate target returns.

3.3.1.2.3 A-Scope - The A-scope shall be provided for measuring receiver sensitivity and lock-on sensitivity of the AN/APG-30 when used in conjunction with the TS-147/UP Signal Generator. The horizontal sweep ranges of 6,000 or 18,000 feet shall permit observation of the corresponding delayed pulses.

(1) A-Scope Calibration - The A scope shall include a calibrated 300 to 600 - cps sawtooth horizontal sweep which may be used to measure the radar repetition rate and the frequency of the variable-frequency 115 V supply.

3.3.1.2.4 Meter Circuits - Specific meter circuits shall be provided to measure the a-c volts, d-c volts, a-c frequency, the preamplifier and AFC crystal current and the magnetron current of the AN/APG-30 Radar Set.

3.3.1.3 Power Input Circuit - The test set shall operate from a primary source of power of 115 V at 320 to 1760 cps. The current requirements shall not be greater than 1.2 amperes.

3.3.1.3.1 Voltage Supply - The power supply shall contain rectifying and voltage regulating circuits to supply 400 V dc, 200 V dc regulated, and -150 V dc regulated.

3.3.1.4 Top Panel Controls and Meters - The following controls and meters, labeled as capitalized, shall be provided on the top panel of the TS-787(*)/APM-72 test set, and shall be edge-lighted in accordance with Specification MIL-P-7788:

(1) DC VOLTS; -250; +150; +300; +200 REF - A four-position rotary switch for selecting the +300, +150, and -250 V dc generated by the AN/APG-30 and the +200 V dc generated by the test set, to be measured by the test set d-c voltmeter.

(2) AC LINE: 400; VAR; OUTPUT OF AC REG - A three-position rotary switch for selecting the 380 to 420-cps and 320 to 1760-cps voltage inputs to the AN/APG-30, and the regulated voltage out-put from the AN/APG-30 a-c regulator, so that they can be measured by the test set a-c frequency meter.

(3) RANGE SELECTOR: 900ft/11V; 2250ft/20 V; 2400ft/21V; 3900ft/31V; 5400ft/40V; 8400ft/61V - A six position rotary switch for selecting the six noted voltages for comparison with radar range

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(3) RANGE SELECTOR: 900ft/11V; 2250ft/20 V; 2400ft/21V; 3900ft/31V; 5400ft/40V; 8400ft/61V - A six position rotary switch for selecting the six noted voltages for comparison with radar range voltage in the test set bridge circuit. The voltages correspond to the noted ranges.

(4) FUNCTION SELECTOR; NORM; SERVO; BRIDGE; ERROR; CAL; ANL - A six position rotary switch for selecting the various functions of the bridge circuit.

(5) FINE BAL - A two-position spring-loaded, normally-open, toggle switch for increasing the sensitivity of the balance meter indication.

(6) BRIDGE BAL - A rotating control for balancing the bridge circuit.

(7) SERVO BAL - A rotating control for adjusting the secondary voltage divider for balance against the precision divider.

(8) DIM - A rotating control for varying the illumination of the edge-lighted panels.

(9) RANGE ZERO - A rotating control for adjusting the RANGE meter to the correct reading.

(10) DC VOLTS - A 4-1/2 inch diameter meter for indicating radar set and test set d-c power supply voltages. The range of the meter shall be from 150 to 300 V. The scale shall have 5-V increment markings with the 150, 200, 250, and 300 points indicated and shall have an accuracy ±1/2 percent of full scale deflection.

(11) AC LINE - A 3-1/2 inch diameter meter with suppressed zero, for indicating the variable frequency input voltage, the 380- to 420-cps line voltage, and the output voltage from the radar a-c regulator. The range of the meter shall be from 80 to 130 V. The scale shall have 1-V increment markings beginning with the 80, 90, 100, 110, 120, and 130 points indicated. VOLTS AC shall appear on the scale face and shall have an accuracy within ±2 percent.

(12) FREQUENCY - CPS - 90-140 V - A 3-1/2 inch diameter meter for indicating the frequency of the 380- to 420-cps input. The range of the meter shall be from 350 to 450 cps. The scale shall have 2-cps increment markings and the 350, 380, 400, 420, and 450 points indicated and shall have an accuracy within 0.4 percent.

(13) UNDER OVER - YARDS ERROR - A 1.5 inch diameter meter, zero center, for indicating unbalance of the bridge circuit in yards. The range shall be ±50 yards. The scale shall have 5-yard increment markings and the 50, 25, 0, 25, and 50 points indicated and shall have an accuracy within ± 3 percent.

(14) BALANCE - A 1-1/2 inch diameter, zero center, meter for indicating unbalance of the bridge circuit. The meter shall have a ± 10 microampere movement and the center balance point shall be indicated.

(15) RANGE - FEET X 1000 - A 1-1/2-inch diameter meter for indicating the range of a signal upon which the radar is locked. The range of the meter shall be from 0 to 12,000 feet. The scale shall have the 0, 2, 4, 6, 8, 10, and 12 thousand points called out. The 900-, 2250-, and 5400-foot points shall be indicated by a smaller scale marking. The meter shall have an accuracy within ± 3 percent.

3.3.1.5 Front Panel Controls, Meters, and Connections - The following controls, meters, and connections, labeled as capitalized, shall be provided on the front panel of the TS-787(*)/APM-72 test set and shall be edge-lighted in accordance with Specification MIL-P-7788;

(1) VERT INPUT; VID; TRIG; 400 LINE; VAR LINE - A four-position rotary switch for selecting the markers produced by the test set.

(2) MARKER SELECTOR 900-2250; 900-5400; TRACK; OFF - A four-position rotary switch for selecting the markers produced by the test set.

(3) SWEEP SELECTOR; 6000 FT; 18,000 FT; VAR - A three-position rotary switch for selection of the sweep length for the test set oscilloscope.

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(4) VARIABLE FREQUENCY CYCLES: 300; 400; 600 - A rotating control for varying the sweep rate. The scale shall have 25-cycle increment markings from 300 to 400 and 50-cycles increment markings from 400 to 600.

(5) +200 V REF ADJ - A rotating control for adjusting the +200 -V level of the test set internal power supply.

(6) POWER: ON; OFF - A two-position toggle switch for turning the test set power on and off.

(7) GATES OUT - A two-position, spring-loaded, normally-open toggle switch for energizing the radar gates-out circuit.

(8) HOR GAIN, HOR POS, VERT GAIN, VERT POS, FOCUS, INTENSITY - Six seperate rotating controls for adjusting the horizontal gain and positioning, vertical gain and positioning, intensity, and focus of the test set oscilloscope.

(9) TRACK RANGE - A rotating control for varying the track marker in range from approximately 900 to 5400 feet.

(10) 28 V TEST - A two-position, spring-loaded, normally-off, toggle switch for applying current from the 28-V source to the magnetron meter for measurement of the 28-V level. The arrow shall point to a line on the meter scale at the 28-V Test Point.

(11) PRE-AMP XTAL MA: AFC XTAL MA - Two 1-1/2 inch diameter meter for indicating the radar preamplifier and afc crystal currents. The range of each meter shall be from 0 to 1 ma. The scale shall have 0.05-ma increment markings, with the 0, 0.2, 0.4, 0.6, 0.8, and 1 points indicated. This meter shall have an accuracy within ±3% and an internal impedance not greater than 150 ohms.

(12) MAG CUR MA - A 1-1/2-inch diameter meter for indicating radar magnetron current and for indicating the 28-V supply when the 28 V TEST switch is engaged. The range of the meter shall be from 0 to 5 ma and 0 to 50 V dc. The scale shall have 0.2 ma increment markings with the 0, 1, 2, 3, 4, and 5 points indicated. The 28 V point shall also be indicated and pointed to by the arrow from the 28 V TEST switch. This meter shall have an accuracy within ±3% and internal impedance not exceeding 30 ohms.

(13) SPARE: 8 AMP - Two fuse holders one for the 8 amp 115-V a-c fuse in the power supply circuit and one for an equivalent spare fuse.

(14) JUNCTION BOX - An AN3102-28-11P receptacle to mate with the AN3106A-28-11S plug on the cable assembly CX-2451/U. A removable cap with captive chain shall be provided for the receptacle.

(15) 115 VOLTS 320-1760 CPS 8 AMP MAX - A Navy type 49435A receptacle, a fused power outlet to provide power for auxiliary test equipment. A removable cap with captive chain shall be provided for the receptacle.

(16) VIDEO: MARKER: ANL: TRIG: PREAMP - Five receptacles UG-290/U to mate with plugs UG-260 on cables carrying the video signal from the range computer power supply, test set markers to the range computer power supply, ANL voltage from the range computer power supply, trigger signal from the radar frequency converter transmitter, and preamp crystal current from the radar frequency converter, respectively.

3.3.1.6 Internal Controls - A control for positioning and/or adjusting each of the following items shall be provided inside the case of the TS-787(*)/APM-72:

(1) Fixed 900-ft video marker

(2) Fixed 2250-ft video marker

(3) Fixed 5400-ft video marker

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(4) Compensate for effect on the 900-ft circuit by switching from the 900-5400 to the 900-2250-ft position

(5) Range of variable video marker

(6) Sweep frequency of the 6000-ft sweep on the oscilloscope

(7) Sweep frequency of the 18,000-ft sweep on the oscilloscope

(8) Calibrate the YARDS ERROR meter

(9) Oscilloscope horizontal sweep sawtooth amplitude

(10) Astigmatic affect on the oscilloscope presentation

(11) Oscilloscope horizontal and vertical linearity

3.3.2 Interconnecting Box J-565/APM-72 - An interconnecting box J-565/APM-72 shall be provided for interconnection of the TS-787(*)/APM-72 and the AN/APG-30. The unit shall consist of:

3.3.2.1 Box - The dimensions of the box shall be as small as possible but not be greater than 8 inches long, 6 inches wide, and 3-1/2 inches high.

3.3.2.2 Connectors - One AN3102-28-11S and one AN3102A-28-17S connector for attaching cable assembly CX-2451/U, and a cable assembly from radar range servo, respectively.

3.3.2.3 Terminal Block Assemblies - Two terminal block assemblies AN3436-1-12 or equivalent, for attaching the cable assembly CX-2452/U.

3.3.2.4 Relays - One relay, contact rating 3 amp/32-V dc and 3 amp/125-V ac.

3.3.2.5 Cable Assembly CX-2452/U - A cable assembly 3 feet long shall be provided as a part of the interconnecting box J-565/APM72. One end shall be made up as pigtails attached in the box and the other end shall terminate in a AN-3106A-28-17P connector, one two prong a-c plug and two phone jack plugs PJ-055B. The part of the cable containing jack plug P-2804 shall be 5 inches long.

3.3.3 Cable Assembly, CX-2451/U - One cable assembly, CX-2451/U, 10 feet long shall be provided with connector AN3106A-28-11S on one end and connector AN3106A-28-11P on the other.

3.3.4 Cable Assembly CG-572/U - One cable assembly, rf, 13 feet long shall be provided consisting of four lengths of RG-71/U cable with UG-260/U connectors on each end, except one which shall be terminated in a phone jack, PJ-005 B.

3.3.5 ANL Probe - One probe shall be provided consisting of 18 inches of RG-71/U connector on one end and a 5 inch probe on the other.

3.3.6 Video Adapter - A video adapter shall be provided as a part of video cable assembly CG-572/U, connecting P-2704 to J-809. The adapter shall be 3-1/16 inches long, 1-1/8 inches wide, and 1-3/16 inches deep. One end of the adapter shall have a UG-604/U receptacle and the other end shall have a 6 inch length of RG-71/U cable terminating in a UG-260/U connector.

3.3.7 Transit Case - A case shall be provided to carry the complete equipment, accessories and Handbook Maintenance Instructions. The dimensions of the case shall not be greater than 26 inches long, 16 inches wide and 16 inches high. The weight of the transit case shall not be greater than 30 lbs.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification - The equipment specified herein shall be subject to preproduction and inspection tests as follows:

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(1) Preproduction Tests - Preproduction tests shall be made on a sample(s) representative of the production equipments, to determine that the production equipments will meet the requirements of this specification and the applicable subsidiary specifications.

(2) Inspection Tests - Inspection tests shall be made on production test sets submitted for acceptance under the contract to determine conformance with the requirements of this specification and the applicable subsidiary specifications and to determine that the test sets are equivalent in performance and construction to the approved preproduction sample. Any production prior to approval of the preproduction sample shall be at the contractor's risk.

4.2 Preproduction Tests - Preproduction tests shall be conducted on two preproduction models representative of the production equipment as follows:

4.2.1 Scope of Preproduction Tests - Preproduction tests shall be conducted at a laboratory designated by the procuring agency, and shall be in accordance with the requirements of Specification MIL-T-945. The procuring agency, however, reserves the right to make additional preproduction tests other than those outlined in Specification MIL-T-945 to insure compliance with this specification and the contract.

4.2.2 Presubmission Tests - Prior to the submission of any models for preproduction testing, the contractor shall furnish facilities for and shall conduct presubmission tests as required in Specification MIL-T-945. In addition, the design data required by Specification MIL-T-945 shall be submitted to the procuring agency for review and approval prior to submission of any models.

4.2.3 Approval of Preproduction Models - Approval of preproduction model shall be by the procuring agency upon satisfactory completion of all tests. No additional models shall be delivered prior to the approval of the preproduction model. Prefabrication of any equipments prior to the approval of the preproduction model is at the contractor's own risk. One approved preproduction model will be returned to the contractor for his use in the fabrication and testing of the models to be submitted for acceptance under the contract. The preproduction model shall not be considered as one of the models under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production model.

4.3 Inspection Tests - The contractor shall furnish all models and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the cognizant Naval Inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the procuring agency. The contractor shall furnish test reports in duplicate, showing quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.3.1 Scope of Inspection Tests - Inspection tests shall meet all the requirements of MIL-T-945, and shall be of the following two types:

(1) Individual Tests - Individual tests shall be conducted on each model submitted for acceptance under the contract.

(2) Sampling Tests - Sampling tests shall be conducted on a quantity as approved by the procuring agency. Models submitted for sampling tests shall be selected by the Government Inspector and shall first have passed the individual tests. Sampling tests shall be more extensive than the individual tests and may include any of the tests listed under preproduction tests which are deemed necessary by the inspector to determine that production models continue to be equivalent in performance and construction of the approved preproduction models.

4.4 Procedures - The procedures for conducting presubmission testing of preproduction models and inspection testing of production models shall be determined by the contractor. These test procedures shall be subject to review and approval by the procuring agency prior to the initiation of any testing.

4.5 Acceptance of Production Models - Acceptance of all production models shall be by the cognizant Naval inspector.

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5. PACKAGING AND PREPARATION FOR SHIPMENT

5.1 Packaging - The equipment and spare parts shall be preserved, packaged, and packed in accordance with the requirements of Specification JAN-P-658 and shall be marked in accordance with the requirements of Specification MIL-STD-129. Suitable interior packing, blocking, and bracing shall be used to prevent shifting in the exterior shipping container and shock from handling in transit.

6. NOTES

6.1 Ordering Information - Invitation for bids and contracts or orders shall specify the title, number, and date of this specification.

PATENT NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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